

Computational Biology and Chemistry

Dear Editors,

Please consider "Junction and register choices in a computational catalogue of NR4A3 fusion gapmer designs" as a Full-length research article in Computational Biology and Chemistry.

This computational catalogue asks how exact RNA-junction and register choices change the sequence matches relevant to designing NR4A3 fusion gapmers. It records 190 designs across 38 modelled junctions. A one-base register change produces an eleven-base normal-NR4A3 match through the DNA gap; a different acceptor requires a different sequence. These are concrete design distinctions, not measurements of biological selectivity.

The adopted ten-base criterion classifies 87 designs as matching mature parent RNA, but artificial exon joins meet the same criterion at a similar rate. We retain this negative control and the cutoff sensitivity without claiming an EMC-specific liability excess. Most modelled junctions are unreported in patients. The released sequence records support conditional design inspection; no cleavage, potency, delivery, safety or therapeutic window has been measured.

The article fits the journal through its computational design and analysis of nucleic-acid interactions. The manuscript includes the full author declarations. The research was self-funded, with no external funding or competing financial interests. The author is an EMC survivor, a declared non-financial interest.

Earlier versions are public on Qeios, including version 4 (<https://www.qeios.com/read/VL3LJR.4>). The historical archive at doi:10.5281/zenodo.22229096 contains the preceding analysis, not all corrected interpretations in this manuscript. Supplementary Files 1 and 2 supply the sequence record and correction note.

Claude and GPT-6-Astra assisted with analysis and manuscript preparation as detailed in the manuscript. The author directed the work and is responsible for its content. We request the standard subscription route and online-only colour.

Sincerely,

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